

8.12 Washer Method: Revolve Around Other Axes

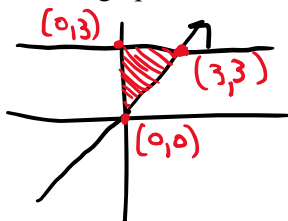
Calculus

Solutions

Practice

1. A region S is bounded by the graphs of $y = x$, $x = 0$, and $y = 3$.

a. Sketch the graph and find the area of region S .



$$A = \int_0^3 (3-x) dx = 4.5$$

b. Let S be the base of a solid with cross sections perpendicular to the x -axis that form a square. Find the volume of this solid. [Use a calculator after you set up the integral.]

$$V = \int_0^3 (3-x)^2 dx$$

$$V = 9$$

c. Let S be the base of a solid with cross sections perpendicular to the y -axis that form a semi-circle. Find the volume of this solid. [Use a calculator after you set up the integral.]

$$V = \int_0^3 \frac{\pi}{2} \left(\frac{y}{2}\right)^2 dy$$

$$V = 3.534$$

Write the equation for the “big radius” and the “little radius” for the solid of revolution when revolving S around the given line. Then setup the integral to find the volume of the solid formed. **DO NOT EVALUATE.**

d. The line $x = 3$.

$$R = 3$$

$$r = 3-y$$

$$V = \pi \int_0^3 [9 - (3-y)^2] dy$$

e. The line $y = -1$.

$$R = 4$$

$$r = x+1$$

$$V = \pi \int_0^3 [16 - (x+1)^2] dx$$

f. The line $x = -1$.

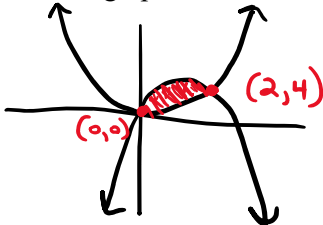
$$R = y+1$$

$$r = 1$$

$$V = \pi \int_0^3 [(y+1)^2 - 1] dy$$

2. A region T is bounded by the graphs of $y = x^2$ and $y = 4x - x^2$.

a. Sketch the graph and find the area of region T .



$$\begin{aligned} x^2 &= 4x - x^2 \\ 2x^2 - 4x &= 0 \\ 2x(x-2) &= 0 \\ x &= 0 \quad x = 2 \end{aligned}$$

$$A = \int_0^2 (4x - x^2) - x^2 dx$$

$$A = \int_0^2 (4x - 2x^2) dx$$

$$A = 2.667$$

b. Let T be the base of a solid with cross sections perpendicular to the x -axis that form a semicircle. Find the volume of this solid. [Use a calculator after you set up the integral.]

$$V = \int_0^2 \frac{\pi}{2} \left(\frac{4x - 2x^2}{2}\right)^2 dx$$

$$V \approx 1.6755$$

Write the equation for the "big radius" and the "little radius" for the solid of revolution when revolving T around the given line. Then setup the integral to find the volume of the solid formed. **DO NOT EVALUATE.**

c. The line $y = 6$.

$$R = 6 - x^2$$

$$r = 6 - (4x - x^2) = x^2 - 4x + 6$$

$$V = \pi \int_0^2 [(6 - x^2)^2 - (x^2 - 4x + 6)^2] dx$$

d. The line $y = -3$.

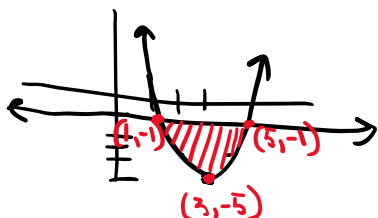
$$R = 3 + 4x - x^2$$

$$r = 3 + x^2$$

$$V = \pi \int_0^2 [(3 + 4x - x^2)^2 - (3 + x^2)^2] dx$$

3. A region D is bounded by the graphs of $y = (x - 3)^2 - 5$ and $y = -1$.

a. Sketch the graph and find the area of region D .



$$\pm\sqrt{y+5} = x-3$$

$$x = 3 \pm \sqrt{y+5}$$

$$y = x^2 - 6x + 9 - 5$$

$$y = x^2 - 6x + 4$$

$$A = \int_1^5 [(-1) - (x^2 - 6x + 4)] dx$$

$$A = \int_1^5 (-x^2 + 6x - 5) dx$$

$$A = 10.667$$

b. Let D be the base of a solid with cross sections perpendicular to the x -axis that form a rectangle with a height 3 times the width. Find the volume of this solid. [Use a calculator after you set up the integral.]

$$V = \int_1^5 [3(-x^2 + 6x - 5)^2] dx$$

$$V = 102.4$$

c. Let D be the base of a solid with cross sections perpendicular to the y -axis that form an isosceles right triangle. Find the volume of this solid. [Use a calculator after you set up the integral.]

$$V = \int_{-5}^{-1} \frac{1}{2} [(3 + \sqrt{y+5}) - (3 - \sqrt{y+5})]^2 dy$$

$$V = \int_{-5}^{-1} \frac{1}{2} (2\sqrt{y+5})^2 dy$$

$$V = 16$$

Write the equation for the "big radius" and the "little radius" for the solid of revolution when revolving D around the given line. Then setup the integral to find the volume of the solid formed. **DO NOT EVALUATE.**

d. The line $y = 9$.

$$R = x^2 - 6x + 4 - 9$$

$$r = 10$$

$$V = \pi \int_1^5 [(x^2 - 6x - 5)^2 - (100)] dx$$

e. The line $x = -2$.

$$R = 3 + \sqrt{y+5} + 2$$

$$r = 3 - \sqrt{y+5} + 2$$

$$V = \pi \int_{-5}^{-1} [(5 + \sqrt{y+5})^2 - (5 - \sqrt{y+5})^2] dy$$

f. The line $x = 5$.

$$R = 5 - (3 - \sqrt{y+5})$$

$$r = 5 - (3 + \sqrt{y+5})$$

$$V = \pi \int_{-5}^{-1} [(2 + \sqrt{y+5})^2 - (2 - \sqrt{y+5})^2] dy$$